

Stage 1: Data acquisition

Google Earth Engine platform

- Climate (5): ERA5-Land
- ▲ Surface (4): Sentinel-2
- Urban 2D (5): Dynamic World
- Urban 3D (4): GHSL + 3D-GloBFF
- Topographic (5): ALOS AW3D30

LST <- Landsat 8/9 ST_B10
Mono-Window, 2019-2021 JJA median
Validation: TRIMS LST 1 km, ERA5 Tair

Stage 2: ROI definition

GHSL SMOD 2020 urban-center
(class 30) + 15 km buffer

41 prefecture-level cities
Hubei 13 / Hunan 14 / Jiangxi 14

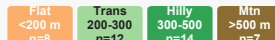
3 concentric zones / ROI:
core <5 km / suburb 5-10 km
peri-urban 10-15 km

250 m analysis grid
Cleaning drops 14.1 %
Final: 1,190,852 cells

Stage 3: Topographic stratification

Cut ROI-mean DEM at
200 / 300 / 500 m

=> Four strata:



Stage 4: Feature engineering

23 drivers in 5 categories

VIF diagnosis (cut-off VIF<5)
13 drivers fail (FAR/BVD inf,
Tair/Tmax ~ 5e5)

Dual feature-set strategy:
full 23 -> tree models
(CatBoost, RF)
VIF-10 -> linear models
(OLS, SLM)

Stage 5: Four complementary models

OLS
linear
R2=0.631

Random Forest
R2=0.811

CatBoost-SHAP
(main model)
R2=0.786

Spatial Lag
SLM, k=8 NN
pseudo-R2=0.717

Validation

Three-layer CV:
(i) 5-fold random CV
(ii) 41-fold LOCO
(iii) within-stratum LOCO

Cross-checks:
TRIMS LST 1 km
ERA5-Land 2 m Tair

Stage 6: Mechanism inference

6a SHAP attribution

mean |SHAP| at 3 scales:
global / per-stratum / per-city
Cell-level SHAP geography (250 m)
Cross-model agreement (CB vs RF)

6b Partial dependence

40-point grid (2nd-98th pct)
Inflection at max |second-diff|
=> stratum-specific operating
thresholds (NDVI, LAI, DEM,
ISA)

6c Piecewise SEM

semopy, Lefcheck (2016) framework
One SEM fit per stratum
4 direct + 3 mediation edges
Fit: CFI / TLI / RMSEA / SRMR

6d Counterfactual UGI

S0 baseline / S1 uniform NDVI+0.30
S2 targeted-adaptive (stratum thr.)
S3 aggressive-adaptive (thr.+0.10)
120 k stratified sub-sample;
metric: delta-LST per %NDVI

Stage 7: Robustness suite

(i) Bootstrap SHAP

500 iter, 95 % CI

(ii) Cut-off

±50, ±100 m

(iii) Tree-family

CB / XGB / LGB / RF

(iv) Latitude band

Koppen-zone proxy

(v) Per-city heatmap

41 cities

(vi) Scale 250 m to 2 km

grid coarsening